Team ISEd Tea 002 Intrusion Report

Table of Contents

Overview:.....2
Waf Activity Report:.....2
SEIM Activity Report:.....2
AD:.....2
TICKETS:.....2
MGMT:.....2
ADMIN:.....2
WWW:.....3

Overview:

Reporting on information acquired from 10:00 am to 12:00 pm CST on 23rd 2024.

WAF Activity Report:

Within each document section (for each box). Images from our IDS system will be used as a primary source of information. For consistency, please refer to these sections listed in the table of contents.

```
[21/Feb/2026:11:26:11 -0600] "GET /vehicles/ HTTP/1.1" 200 13775 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:11 -0600] "POST /vehicles/6a53d7be-5742-40d7-97ab-6215bed5d9b5 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:12 -0600] "GET /vehicles/ HTTP/1.1" 200 13790 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:12 -0600] "POST /vehicles/ce83fa1f-786f-4637-8833-daebf1603401 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:12 -0600] "GET /vehicles/ HTTP/1.1" 200 13787 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:12 -0600] "POST /vehicles/e5ac8f47-d163-4abc-9b71-eb155c5794f3 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
- - [21/Feb/2026:11:26:13 -0600] "POST /login HTTP/1.1" 302 189 "http://admin.team7.isucdc.com/login" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) Gecko/20100101 Firefox/1
- - [21/Feb/2026:11:26:13 -0600] "GET / HTTP/1.1" 200 919 "http://admin.team7.isucdc.com/login" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) Gecko/20100101 Firefox/147.0"
- - [21/Feb/2026:11:26:13 -0600] "GET /static/css/style.css HTTP/1.1" 304 0 "http://admin.team7.isucdc.com/" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) Gecko/20100101 Fi
- - [21/Feb/2026:11:26:13 -0600] "GET /static/js/map.js HTTP/1.1" 200 9583 "http://admin.team7.isucdc.com/" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) Gecko/20100101 Fir
- - [21/Feb/2026:11:26:13 -0600] "GET /api/vehicles HTTP/1.1" 200 13786 "http://admin.team7.isucdc.com/" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) Gecko/20100101 Firefo
[21/Feb/2026:11:26:14 -0600] "GET /vehicles/ HTTP/1.1" 200 13786 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:14 -0600] "POST /vehicles/837b1b7e-9fb8-4de5-9654-47d9b5c141e5 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:14 -0600] "GET /vehicles/ HTTP/1.1" 200 13784 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:14 -0600] "POST /vehicles/3e409a45-3b1e-4661-8cd1-1f67737f3f04 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
- - [21/Feb/2026:11:26:14 -0600] "GET /api/vehicles HTTP/1.1" 200 13783 "http://admin.team7.isucdc.com/" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) Gecko/20100101 Firefo
[21/Feb/2026:11:26:14 -0600] "GET /vehicles/ HTTP/1.1" 200 13783 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:15 -0600] "POST /vehicles/c6de83f7-d539-4d40-905a-4fea924fc011 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
- - [21/Feb/2026:11:26:15 -0600] "GET /api/vehicles HTTP/1.1" 200 13787 "http://admin.team7.isucdc.com/" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) Gecko/20100101 Firefo
[21/Feb/2026:11:26:16 -0600] "GET /vehicles/ HTTP/1.1" 200 13787 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:16 -0600] "GET /vehicles/ HTTP/1.1" 200 13787 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:16 -0600] "POST /vehicles/ffac9a5f-ecde-42e8-a3ec-e8f442b79009 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:16 -0600] "POST /vehicles/ae962416-fefc-420d-af02-aa478c8880c4 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
- - [21/Feb/2026:11:26:16 -0600] "GET /api/vehicles HTTP/1.1" 200 13789 "http://admin.team7.isucdc.com/" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) Gecko/20100101 Firefo
[21/Feb/2026:11:26:16 -0600] "GET /vehicles/ HTTP/1.1" 200 13789 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:16 -0600] "GET /vehicles/ HTTP/1.1" 200 13789 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:16 -0600] "GET /vehicles/ HTTP/1.1" 200 13789 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:16 -0600] "POST /vehicles/0ab054ac-778e-4578-8933-d0bcc16701fa HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:16 -0600] "POST /vehicles/d351dac7-43ae-4c46-bad9-f48df35ca5ed HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:17 -0600] "POST /vehicles/c8177914-d1f0-4371-89fe-72a14453b664 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
- - [21/Feb/2026:11:26:17 -0600] "GET /api/vehicles HTTP/1.1" 200 13797 "http://admin.team7.isucdc.com/" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) Gecko/20100101 Firefo
[21/Feb/2026:11:26:17 -0600] "GET /vehicles/ HTTP/1.1" 200 13797 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:17 -0600] "GET /vehicles/ HTTP/1.1" 200 13797 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:17 -0600] "POST /vehicles/387cb109-7057-452a-ae73-b7518e947efc HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:18 -0600] "POST /vehicles/c5f60fcc-fc14-4e13-8989-09e3598b2dcc HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
- - [21/Feb/2026:11:26:18 -0600] "GET /api/vehicles HTTP/1.1" 200 13797 "http://admin.team7.isucdc.com/" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) Gecko/20100101 Firefo
[21/Feb/2026:11:26:18 -0600] "GET /vehicles/ HTTP/1.1" 200 13797 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:18 -0600] "POST /vehicles/837b1b7e-9fb8-4de5-9654-47d9b5c141e5 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:19 -0600] "GET /vehicles/ HTTP/1.1" 200 13794 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:19 -0600] "GET /vehicles/ HTTP/1.1" 200 13805 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:19 -0600] "POST /vehicles/ce83fa1f-786f-4637-8833-daebf1603401 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:19 -0600] "POST /vehicles/6a53d7be-5742-40d7-97ab-6215bed5d9b5 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
- - [21/Feb/2026:11:26:19 -0600] "GET /api/vehicles HTTP/1.1" 200 13803 "http://admin.team7.isucdc.com/" "Mozilla/5.0 (X11; Linux x86_64; rv:147.0) Gecko/20100101 Firefo
[21/Feb/2026:11:26:20 -0600] "GET /vehicles/ HTTP/1.1" 200 13803 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:20 -0600] "GET /vehicles/ HTTP/1.1" 200 13803 "-" "Java-http-client/21.0.9"
[21/Feb/2026:11:26:20 -0600] "POST /vehicles/e5ac8f47-d163-4abc-9b71-eb155c5794f3 HTTP/1.1" 200 43 "-" "Java-http-client/21.0.9"
```

This is the normal traffic we havent seen anything abnormal… this is the typical traffic… we have been saving this log files into backups and this is the typically constant traffic of the communication of the bus system between frontend, backend etc.

```
cdc@WAF:~/waf_logs$ sudo bash -c 'for f in /var/log/nginx/access.log /var/log/nginx/error.log /var/log/modsec_audit.log; do cp "$f" /home/cdc/waf_logs/"$(basename $f).$(date +%H%M%S).bak" && truncate -s 0 "$f" && echo "Done: $f"; done'
Done: /var/log/nginx/access.log
Done: /var/log/nginx/error.log
Done: /var/log/modsec_audit.log
cdc@WAF:~/waf_logs$ ls
access.log.064014.bak          access.log.20260221_0300.bak
access.log.115535.bak          access.log.20260221_0400.bak
access.log.131804.bak          access.log.20260221_0500.bak
access.log.191924.bak          access.log.20260221_0600.bak
access.log.20260220_1400.bak   access.log.20260221_0700.bak
access.log.20260220_1500.bak   access.log.20260221_0800.bak
access.log.20260220_1600.bak   access.log.20260221_0900.bak
access.log.20260220_1700.bak   access.log.20260221_1000.bak
access.log.20260220_1800.bak   access.log.20260221_1100.bak
access.log.20260220_1900.bak   error.log.064015.bak
access.log.20260220_2000.bak   error.log.115535.bak
access.log.20260220_2100.bak   error.log.131804.bak
access.log.20260220_2200.bak   error.log.191924.bak
access.log.20260220_2300.bak   modsec_audit.log.064015.bak
access.log.20260221_0000.bak   modsec_audit.log.115535.bak
access.log.20260221_0100.bak   modsec_audit.log.131804.bak
access.log.20260221_0200.bak   modsec_audit.log.191924.bak
cdc@WAF:~/waf_logs$
```

SEIM Activity Report:

Within each document section (for each box). Images from our IDS system will be used as a primary source of information. For consistency, please refer to these sections listed in the table of contents.

Feb 21, 2026 @ 11:30:08.596 predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 17:29:54 inpu
log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 manager.name: wazuh2 data.dstuser: rog
rule.firedtimes: 34 rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.8, C
7.3 rule.description: PAM: Login session closed. rule.groups: pam, syslog rule.id: 5502 rule.nist_800_53: AU.14, A
rule.gpg13: 7.8, 7.9 rule.gdpr: IV_32.2 location: journald decoder.parent: pam decoder.name: pam id: 1771695008.

Feb 21, 2026 @ 11:30:08.589 predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 17:29:54 inpu
log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 manager.name: wazuh2 data.uid: 0
data.dstuser: roger rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.8, C
7.3 rule.description: PAM: Login session opened. rule.groups: pam, syslog, authentication_success rule.nist_800_53:
C.7 rule.gdpr: IV_32.2 rule.firedtimes: 45 rule.mitre.technique: Valid Accounts rule.mitre.id: T1078 rule.mitre.ta

Feb 21, 2026 @ 11:30:08.559 predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: systemd predecoder.timestamp: Feb 21 17:29:54
input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 manager.name: wazuh2 data.
data.dstuser: roger rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.8, C
7.3 rule.description: PAM: Login session opened. rule.groups: pam, syslog, authentication_success rule.nist_800_53:
C.7 rule.gdpr: IV_32.2 rule.firedtimes: 44 rule.mitre.technique: Valid Accounts rule.mitre.id: T1078 rule.mitre.ta

Feb 21, 2026 @ 11:30:08.542 predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 17:29:53 inpu
log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 data.srcip: 12.110.207.17 data.dstuser:
ox data.srcport: 45204 manager.name: wazuh2 rule.mail: false rule.level: 3 rule.hipaa: 164.312.b rule.pci_dss: 1
rule.tsc: CC6.8, CC7.2, CC7.3 rule.description: sshd: authentication success. rule.groups: syslog, sshd, authentica
ess rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_32.2 rule.firedtimes: 15 rule.mitre.technique: Valid Accounts, Rem

Above screenshot shows totally legit signins. All are domain accounts using the services as expected. This has been the extent of the Wazuh logs for the past while. No signs of intrusion. I have been watching the logs and nothing unusual has come up. I have seen a few cdc or root logins on MGMT, but confirmed it was a teammate SSHing into the box, and also went on and confirmed the only SSH session was from us.

```
[cdc@mgmt ~]$ ss -tp | grep ssh
ESTAB 0      0       192.168.1.30:ssh   192.168.1.25:39834
[cdc@mgmt ~]$ _
```

AD:

We havent seen anything abnormal on this machine to believe it has been broken into. There is nothing special on wireshark, waf, or wazuh for this machine

```
cdc@Ubuntu-External:~$ sudo nmap -sS -p- -Pn ad.team
[sudo] password for cdc:
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-2
Nmap scan report for ad.team7.isucdc.com (49.49.33.1
Host is up (0.0024s latency).
Not shown: 65532 filtered ports
PORT     STATE SERVICE
389/tcp  open  ldap
636/tcp  open  ldapssl
3389/tcp open  ms-wbt-server
MAC Address: BC:24:11:E1:54:D1 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 104.7
cdc@Ubuntu-External:~$
```

```
cdc@WAF:~/waf_logs$ sudo nmap -sS -Pn -p- 192.168.1.
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-2
Nmap scan report for 192.168.1.10
Host is up (0.0023s latency).
Not shown: 65516 filtered ports
PORT      STATE SERVICE
53/tcp    open  domain
88/tcp    open  kerberos-sec
135/tcp   open  msrpc
389/tcp   open  ldap
445/tcp   open  microsoft-ds
464/tcp   open  kpasswd5
593/tcp   open  http-rpc-epmap
636/tcp   open  ldapssl
3268/tcp  open  globalcatLDAP
3269/tcp  open  globalcatLDAPssl
3389/tcp  open  ms-wbt-server
5985/tcp  open  wsman
9389/tcp  open  adws
49666/tcp open  unknown
49667/tcp open  unknown
49679/tcp open  unknown
49680/tcp open  unknown
49705/tcp open  unknown
49746/tcp open  unknown
MAC Address: BC:24:11:48:52:7D (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 104.7
cdc@WAF:~/waf_logs$
```

The nmap ports are the same and nothing has changed…

TICKETS:

We havent seen anything abnormal on this machine to believe it has been broken into. There is nothing special on wireshark, waf, or wazuh for this machine

```
cdc@Ubuntu-External:~$ sudo nmap -sS -p- -Pn tickets
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-2
Nmap scan report for tickets.team7.isucdc.com (49.49
Host is up (0.0013s latency).
Not shown: 65533 filtered ports
PORT     STATE SERVICE
3389/tcp open  ms-wbt-server
5000/tcp open  upnp
MAC Address: BC:24:11:E1:54:D1 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 104.7
```

```
cdc@WAF:~/waf_logs$ sudo nmap -sS -Pn -p- 192.168.1.
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-2
Nmap scan report for 192.168.1.20
Host is up (0.00028s latency).
Not shown: 65523 closed ports
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
3389/tcp  open  ms-wbt-server
5000/tcp  open  upnp
49664/tcp open  unknown
49665/tcp open  unknown
49666/tcp open  unknown
49667/tcp open  unknown
49668/tcp open  unknown
49669/tcp open  unknown
49670/tcp open  unknown
49676/tcp open  unknown
MAC Address: BC:24:11:DC:E2:CB (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 13.09
cdc@WAF:~/waf_logs$
```

The nmap ports are the same and nothing has changed…

MGMT:

```
cdc@Ubuntu-External:~$ sudo nmap -sS -p- -Pn mgmt.te
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-2
Nmap scan report for mgmt.team7.isucdc.com (49.49.33
Host is up (0.00093s latency).
Not shown: 65533 filtered ports
PORT     STATE SERVICE
22/tcp   open  ssh
5000/tcp open  upnp
MAC Address: BC:24:11:E1:54:D1 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 104.5
```

```
cdc@WAF:~/waf_logs$ sudo nmap -sS -Pn -p- 192.168.1.
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-2
Nmap scan report for 192.168.1.30
Host is up (0.00059s latency).
Not shown: 65533 filtered ports
PORT     STATE  SERVICE
22/tcp   open   ssh
5000/tcp closed upnp
MAC Address: BC:24:11:A6:8B:46 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 104.6
```

The nmap ports are the same and nothing has changed…

```
MariaDB [(none)]> use mysql
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
MariaDB [mysql]> show tables;
+---------------------------+
| Tables_in_mysql           |
+---------------------------+
| column_stats              |
| columns_priv              |
| db                        |
| event                     |
| func                      |
| general_log               |
| gtid_slave_pos            |
| help_category             |
| help_keyword              |
| help_relation             |
| help_topic                |
| host                      |
| index_stats               |
| innodb_index_stats        |
| innodb_table_stats        |
| plugin                    |
| proc                      |
| procs_priv                |
| proxies_priv              |
| roles_mapping             |
| servers                   |
| slow_log                  |
| table_stats               |
| tables_priv               |
| time_zone                 |
| time_zone_leap_second     |
| time_zone_name            |
| time_zone_transition      |
| time_zone_transition_type |
| transaction_registry      |
| user                      |
+---------------------------+
31 rows in set (0.001 sec)

MariaDB [mysql]> select * from general_log
    -> ;
Empty set (0.001 sec)

MariaDB [mysql]> _
```

We had logging turned off for our database and we turned it on… SQL has backwards logging so we collected the logs now

Based on the logs we were able to see that the attacker created a SQL user called blue_flag and then using that user was able to collect the flag data using that user.

```
[cdc@mgmt ~]$ mysql -u cdc -pTrueJadeLamb -e "SELECT * FROM payment_info WHERE username='blue_flag';" transit 2>/dev/null
+--------------------------------------+-----------+-------------+-----------------+-------------+-----+
| id                                   | username  | card_number | cardholder_name | expiry_date | cvv |
+--------------------------------------+-----------+-------------+-----------------+-------------+-----+
| 104ff709-e969-4d4f-a493-7f6cb42ab00c | blue_flag | DB_FLAG     | BLUE            | 00/0000     | 000 |
+--------------------------------------+-----------+-------------+-----------------+-------------+-----+
[cdc@mgmt ~]$
```

This is the screenshot of the user.

Also based on the lack of account traffic and activity on the users on the machine… and no lateral movement between the machines on the wireshark we dont believe there was any form of escalated privileges either.

```
Last login: Sat Feb 21 11:00:32 from 49.49.33.15
[cdc@mgmt ~]$ last -a | head -40
cdc      tty1         Sat Feb 21 11:57   still logged in
cdc      pts/2        Sat Feb 21 11:00   still logged in    49.49.33.15
cdc      pts/1        Sat Feb 21 09:45 - 11:11  (01:26)     192.168.1.35
cdc      tty1         Sat Feb 21 09:20 - 11:46  (02:26)
cdc      pts/0        Sat Feb 21 08:27   still logged in    192.168.1.25
cdc      tty1         Sat Feb 21 08:27 - 08:35  (00:08)
cdc      tty1         Sat Feb 21 08:09 - 08:21  (00:12)
cdc      pts/0        Sat Feb 21 08:05 - 08:26  (00:20)     192.168.1.25
melissa. pts/0        Sat Feb 21 07:55 - 07:55  (00:00)     12.110.206.242
cdc      pts/0        Sat Feb 21 07:26 - 07:46  (00:20)     192.168.1.25
cdc      tty1         Sat Feb 21 07:09 - 08:03  (00:53)
cdc      pts/0        Fri Feb 20 16:58 - 00:28  (07:30)     192.168.1.45
cdc      pts/2        Fri Feb 20 14:05 - 00:04  (09:59)     192.168.1.25
cdc      pts/1        Fri Feb 20 13:38 - 17:02  (03:23)     192.168.1.25
cdc      pts/0        Thu Feb 19 14:28 - 16:54 (1+02:25)    192.168.1.50
melissa. pts/0        Thu Feb 19 09:30 - 09:30  (00:00)     12.110.205.163
roger.co pts/0        Thu Feb 19 09:25 - 09:25  (00:00)     68.32.254.50
melissa. pts/0        Thu Feb 19 09:20 - 09:20  (00:00)     12.110.205.162
melissa. pts/0        Thu Feb 19 09:15 - 09:15  (00:00)     68.32.254.50
roger.co pts/0        Thu Feb 19 09:10 - 09:10  (00:00)     12.110.205.161
roger.co pts/0        Thu Feb 19 09:05 - 09:05  (00:00)     68.32.254.50
roger.co pts/0        Thu Feb 19 09:00 - 09:00  (00:00)     12.110.205.160
melissa. pts/0        Thu Feb 19 08:55 - 08:55  (00:00)     68.32.254.50
roger.co pts/0        Thu Feb 19 08:50 - 08:50  (00:00)     12.110.205.159
melissa. pts/0        Thu Feb 19 08:45 - 08:45  (00:00)     68.32.254.50
melissa. pts/0        Thu Feb 19 08:40 - 08:40  (00:00)     12.110.205.158
melissa. pts/0        Thu Feb 19 08:35 - 08:35  (00:00)     68.32.254.50
melissa. pts/0        Thu Feb 19 08:30 - 08:30  (00:00)     12.110.205.157
melissa. pts/0        Thu Feb 19 08:25 - 08:25  (00:00)     68.32.254.50
roger.co pts/0        Thu Feb 19 08:20 - 08:20  (00:00)     12.110.205.156
melissa. pts/0        Thu Feb 19 08:15 - 08:15  (00:00)     68.32.254.50
melissa. pts/0        Thu Feb 19 08:10 - 08:10  (00:00)     12.110.205.155
melissa. pts/0        Thu Feb 19 08:05 - 08:05  (00:00)     68.32.254.50
roger.co pts/0        Thu Feb 19 08:00 - 08:00  (00:00)     12.110.205.154
melissa. pts/0        Thu Feb 19 07:55 - 07:55  (00:00)     68.32.254.50
roger.co pts/0        Thu Feb 19 07:50 - 07:50  (00:00)     12.110.205.153
roger.co pts/0        Thu Feb 19 07:45 - 07:45  (00:00)     68.32.254.50
melissa. pts/0        Thu Feb 19 07:40 - 07:40  (00:00)     12.110.205.152
roger.co pts/0        Thu Feb 19 07:35 - 07:35  (00:00)     68.32.254.50
roger.co pts/0        Thu Feb 19 07:30 - 07:30  (00:00)     12.110.205.151
[cdc@mgmt ~]$ _
```

Based on this traffic we dont think there is any lateral movement. Mgmt doesnt access any other

boxes.

```
[cdc@mgmt ~]$ ss -tp | grep ssh
ESTAB 0      0      192.168.1.30:ssh   192.168.1.25:39834
[cdc@mgmt ~]$ _
```

We checked the users on mgmt and we are the only user on the block

Time	_source
Feb 21, 2026 @ 09:27:30.066	predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sudo predecoder.timestamp: Feb 21 15:27:14 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 manager.name: wazuh2 data.dstuser: root rule.firedtimes: 17 rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.8, CC7.2, CC 7.3 rule.description: PAM: Login session closed. rule.groups: pam, syslog rule.id: 5502 rule.nist_800_53: AU.14, AC.7 rule.gpg13: 7.8, 7.9 rule.gdpr: IV_32.2 location: journald decoder.parent: pam decoder.name: pam id: 1771687650.19118705
Feb 21, 2026 @ 09:27:30.064	predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sudo predecoder.timestamp: Feb 21 15:27:14 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 manager.name: wazuh2 data.srcuser: cdc data.uid: 1000 data.dstuser: root rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC 6.8, CC7.2, CC7.3 rule.description: PAM: Login session opened. rule.groups: pam, syslog, authentication_success rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_32.2 rule.firedtimes: 28 rule.mitre.technique: Valid Accounts rule.mitre.id: T
Feb 21, 2026 @ 09:27:30.064	predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sudo predecoder.timestamp: Feb 21 15:27:14 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 manager.name: wazuh2 data.srcuser: cdc data.dstuser: root data.tty: pts/0 data.pwd: /opt/backend-mgmt/app data.command: /bin/journalctl -u backend --since '10 min utes ago' rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5, 10.2.2 rule.hipaa: 164.312.b rule.tsc: CC6.8, CC7.2, CC7.3 rule.description: Successful sudo to ROOT executed. rule.groups: syslog, sudo rule.nist_800_53: AU.14, AC.7, AC.6
	U.14, AC.7 rule.gdpr: IV_32.2 rule.firedtimes: 19 rule.mitre.technique: Valid Accounts, Remote Services rule.mitre.id: T1078, T1021
Feb 21, 2026 @ 07:39:46.985	predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 13:39:31 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 data.srcip: 12.110.206.206 data.dstuser: root data.srcport: 52900 manager.name: wazuh2 rule.mail: false rule.level: 5 rule.hipaa: 164.312.b rule.pci_dss: 10.2.4, 10.2.5 rule.tsc: CC6.1, CC6.8, CC7.2, CC7.3 rule.description: sshd: authentication failed. rule.groups: syslog, sshd, authentication_failed rule.nist_800_53: AU.14, AC.7 rule.gdpr: IV_35.7.d, IV_32.2 rule.firedtimes: 17 rule.mitre.technique: Password Guessing, SSH

Feb 21, 2026 @ 07:05 predecoder.hostname: wazuh2 predecoder.program_name: kernel predecoder.timestamp: Feb 21 13:05:27 input.type: log agent.name: wazuh 2 agent.id: 000 manager.name: wazuh2 data.extra_data: open data.status: DENIED rule.firedtimes: 34 rule.mail: false rule.level: 3 rule.description: Apparmor DENIED rule.groups: local, syslog, apparmor rule.id: 52002 location: journald decoder.parent: kernel decoder.name: kernel id: 1771679128.14528034 full_log: Feb 21 13:05:27 wazuh2 kernel: audit: type=1400 audit(1771679127.304:362): app armor="DENIED" operation="open" class="file" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=6729 comm="MemoryPoller" r

Feb 21, 2026 @ 07:05:17.931 predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: systemd predecoder.timestamp: Feb 21 13:05:02 input.type: log agent.ip: 192.168.1.30 agent.name: mgmt.team7.isucdc.com agent.id: 005 manager.name: wazuh2 data.dstuser: roger rule.firedtimes: 7 rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.8, CC7.2, CC7.3 rule.description: PAM: Log in session closed. rule.groups: pam, syslog rule.id: 5502 rule.nist_800_53: AU.14, AC.7 rule.gpg13: 7.8, 7.9 rule.gdpr: IV_32.2 location: journald decoder.parent: pam decoder.name: pam id: 1771679117.14527614 full_log: Feb 21 13:05:02 mgmt.team7.isucdc.com sy

Feb 21, 2026 @ 07:05:12.807 predecoder.hostname: wazuh2 predecoder.program_name: kernel predecoder.timestamp: Feb 21 13:05:11 input.type: log agent.name: wazuh 2 agent.id: 000 manager.name: wazuh2 data.extra_data: open data.status: DENIED rule.firedtimes: 33 rule.mail: false rule.level: 3 rule.description: Apparmor DENIED rule.groups: local, syslog, apparmor rule.id: 52002 location: journald decoder.parent: kernel decoder.name: kernel id: 1771679112.14527212 full_log: Feb 21 13:05:11 wazuh2 kernel: audit: type=1400 audit(1771679111.296:361): app armor="DENIED" operation="open" class="file" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=6729 comm="MemoryPoller" r

Feb 21, 2026 @ 07:05:06.806 predecoder.hostname: wazuh2 predecoder.program_name: kernel predecoder.timestamp: Feb 21 13:05:06 input.type: log agent.name: wazuh 2 agent.id: 000 manager.name: wazuh2 data.extra_data: open data.status: DENIED rule.firedtimes: 32 rule.mail: false rule.level: 3 rule.description: Apparmor DENIED rule.groups: local, syslog, apparmor rule.id: 52002 location: journald decoder.parent: kernel decoder.name: kernel id: 1771679106.14526810 full_log: Feb 21 13:05:06 wazuh2 kernel: audit: type=1400 audit(1771679106.164:360): app armor="DENIED" operation="open" class="file" profile="snap.firefox.firefox" name="/proc/pressure/memory" pid=6729 comm="MemoryPoller" r

Feb 21, 2026 @ 07:05:06.577 predecoder.hostname: www.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 13:04:54 input.type: log agent.ip: 192.168.1.50 agent.name: www.team7.isucdc.com agent.id: 004 manager.name: wazuh2 data.dstuser: melissa rule.firedtimes: 6 rule.mail: false rule.level: 3 rule.pci_dss: 10.2.5 rule.hipaa: 164.312.b rule.tsc: CC6.8, CC7.2, CC7.3 rule.description: PAM: L ogin session closed. rule.groups: pam, syslog rule.id: 5502 rule.nist_800_53: AU.14, AC.7 rule.gpg13: 7.8, 7.9 rule.gdpr: IV_32.2 location: journald decoder.parent: pam decoder.name: pam id: 1771679106.14526398 full_log: Feb 21 13:04:54 www.team7.isucdc.com ssh

Feb 21, 2026 @ 07:05:06.034 predecoder.hostname: mgmt.team7.isucdc.com predecoder.program_name: sshd predecoder.timestamp: Feb 21 13:04:51 input.type: log

Roger.cox logged on. But we didnt see any malicious traffic for this user

```
POST /auth/login HTTP/1.1
Upgrade-Insecure-Requests: 1
Cache-Control: no-cache
Accept: */*
Pragma: no-cache
Connection: keep-alive
Accept-Encoding: gzip, deflate
Content-Length: 59
Priority: u=0, i
Accept-Language: en-US,en;q=0.9
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:147.0) Gecko/20100101 Firefox/147.0
Host: mgmt.team7.isucdc.com:5000
Origin: http://mgmt.team7.isucdc.com:5000

---LXcKMrSP---D--

---LXcKMrSP---E--
<html>\x0d\x0a<head><title>302 Found</title></head>\x0d\x0a<body>\x0d\x0a<center><h1>302 Found</h1></center>\x0d\x0a<hr><center>nginx/1.18.0 (Ubuntu)</center>\x0d\x0a</body>\x0d\x0a</html>\x0d\x0a

---LXcKMrSP---F--
HTTP/1.1 302
Server: nginx/1.18.0
Date: Sat, 21 Feb 2026 15:26:46 GMT
Content-Length: 154
Content-Type: text/html
Connection: keep-alive
Location: /auth/login

---LXcKMrSP---H--
ModSecurity: Warning. Matched "Operator `Eq' with parameter `0' against variable `REQUEST_HEADERS:Content-Type' (Value: `0' ) [file "/usr/local/modsecurity/coreruleset/rules/REQUEST-920-PROTOCOL-ENFORCEME
NT.conf"] [line "713"] [id "920340"] [rev ""] [msg "Content-Type header missing from request with non-zero Content-Length"] [data ""] [severity "2"] [ver "OWASP_CRS/4.24.0-dev"] [maturity "0"] [accuracy "
0"] [tag "application-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-protocol"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "OWASP_CRS/PROTOCOL-ENFORCEMENT"] [tag "capec/1000/210/2
72"] [hostname "mgmt.team7.isucdc.com"] [uri "/auth/login"] [unique_id "177168760631.604568"] [ref "v357,2"]
ModSecurity: Warning. Matched "Operator `Eq' with parameter `0' against variable `REQUEST_HEADERS:Content-Type' (Value: `0' ) [file "/usr/local/modsecurity/coreruleset/rules/REQUEST-920-PROTOCOL-ENFORCEME
NT.conf"] [line "1082"] [id "920640"] [rev ""] [msg "Content-Type header missing from request with body"] [data ""] [severity "2"] [ver "OWASP_CRS/4.24.0-dev"] [maturity "0"] [accuracy "0"] [tag "applicat
ion-multi"] [tag "language-multi"] [tag "platform-multi"] [tag "attack-protocol"] [tag "paranoia-level/1"] [tag "OWASP_CRS"] [tag "OWASP_CRS/PROTOCOL-ENFORCEMENT"] [tag "capec/1000/210/272"] [hostname "mg
mt.team7.isucdc.com"] [uri "/auth/login"] [unique_id "177168760631.604568"] [ref "v402,59"]
ModSecurity: Access denied with code 302 (phase 2). Matched "Operator `Ge' with parameter `5' against variable `TX:BLOCKING_INBOUND_ANOMALY_SCORE' (Value: `10' ) [file "/usr/local/modsecurity/coreruleset/
rules/REQUEST-949-BLOCKING-EVALUATION.conf"] [line "222"] [id "949110"] [rev ""] [msg "Inbound Anomaly Score Exceeded (Total Score: 10)"] [data ""] [severity "0"] [ver "OWASP_CRS/4.24.0-dev"] [maturity "0
"] [accuracy "0"] [tag "anomaly-evaluation"] [tag "OWASP_CRS"] [hostname "mgmt.team7.isucdc.com"] [uri "/auth/login"] [unique_id "177168760631.604568"] [ref ""]

---LXcKMrSP---I--

---LXcKMrSP---J--

---LXcKMrSP---Z--


==> /var/log/nginx/access.log <==
12.110.206.246 - - [21/Feb/2026:09:26:46 -0600] "GET /auth/login HTTP/1.1" 405 153 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10.15; rv:147.0) Gecko/20100101 Firefox/147.0"
```

We also saw around 9:35 an blocked injection by our WAF

```
Feb 21 09:28:31 mgmt.team7.isucdc.com sshd[1187168]: pam_sss(sshd:auth): authentication success; logname= uid=0 euid=0 tty=ssh >
Feb 21 09:28:32 mgmt.team7.isucdc.com sshd[1187168]: Accepted password for krystal.gray from 12.110.206.207 port 38414 ssh2
Feb 21 09:28:32 mgmt.team7.isucdc.com sshd[1187168]: pam_unix(sshd:session): session opened for user krystal.gray by (uid=0)
Feb 21 09:28:32 mgmt.team7.isucdc.com sshd[1187168]: pam_unix(sshd:session): session closed for user krystal.gray
Feb 21 09:28:33 mgmt.team7.isucdc.com sshd[1187201]: pam_sss(sshd:auth): authentication success; logname= uid=0 euid=0 tty=ssh >
Feb 21 09:28:34 mgmt.team7.isucdc.com sshd[1187201]: Accepted password for joann.wilson from 12.110.206.207 port 46322 ssh2
Feb 21 09:28:35 mgmt.team7.isucdc.com sshd[1187230]: pam_sss(sshd:auth): authentication success; logname= uid=0 euid=0 tty=ssh >
Feb 21 09:28:36 mgmt.team7.isucdc.com sshd[1187230]: Accepted password for leslie.brady from 12.110.206.207 port 46326 ssh2
Feb 21 09:29:48 mgmt.team7.isucdc.com sshd[1187642]: pam_sss(sshd:auth): authentication success; logname= uid=0 euid=0 tty=ssh >
Feb 21 09:29:49 mgmt.team7.isucdc.com sshd[1187642]: Accepted password for roger.cox from 12.110.207.3 port 33660 ssh2
Feb 21 09:29:49 mgmt.team7.isucdc.com sshd[1187642]: pam_unix(sshd:session): session opened for user roger.cox by (uid=0)
Feb 21 09:34:49 mgmt.team7.isucdc.com sshd[1189114]: pam_sss(sshd:auth): authentication success; logname= uid=0 euid=0 tty=ssh >
Feb 21 09:34:51 mgmt.team7.isucdc.com sshd[1189114]: Accepted password for melissa.robles from 68.32.254.50 port 36614 ssh2
Feb 21 09:34:51 mgmt.team7.isucdc.com sshd[1189114]: pam_unix(sshd:session): session opened for user melissa.robles by (uid=0)
Feb 21 09:34:51 mgmt.team7.isucdc.com sshd[1189114]: pam_unix(sshd:session): session closed for user melissa.robles
Feb 21 09:39:48 mgmt.team7.isucdc.com sshd[1190602]: pam_sss(sshd:auth): authentication success; logname= uid=0 euid=0 tty=ssh >
Feb 21 09:39:51 mgmt.team7.isucdc.com sshd[1190602]: Accepted password for melissa.robles from 12.110.207.4 port 46918 ssh2
Feb 21 09:39:51 mgmt.team7.isucdc.com sshd[1190602]: pam_unix(sshd:session): session opened for user melissa.robles by (uid=0)
Feb 21 09:39:51 mgmt.team7.isucdc.com sshd[1190602]: pam_unix(sshd:session): session closed for user melissa.robles
Feb 21 09:43:03 mgmt.team7.isucdc.com sshd[1191584]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh>
Feb 21 09:43:05 mgmt.team7.isucdc.com sshd[1191584]: Failed password for root from 49.10.219.164 port 35546 ssh2
Feb 21 09:43:06 mgmt.team7.isucdc.com sshd[1191584]: error: maximum authentication attempts exceeded for root from 49.10.219.16>
Feb 21 09:43:06 mgmt.team7.isucdc.com sshd[1191584]: Disconnecting authenticating user root 49.10.219.164 port 35546: Too many >
Feb 21 09:43:06 mgmt.team7.isucdc.com sshd[1191607]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh>
Feb 21 09:43:06 mgmt.team7.isucdc.com sshd[1191607]: pam_sss(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh >
Feb 21 09:43:06 mgmt.team7.isucdc.com sshd[1191607]: pam_sss(sshd:auth): received for user cdc: 10 (User not known to the under>
Feb 21 09:43:08 mgmt.team7.isucdc.com sshd[1191607]: Failed password for cdc from 49.10.219.164 port 35554 ssh2
Feb 21 09:43:08 mgmt.team7.isucdc.com sshd[1191607]: error: maximum authentication attempts exceeded for cdc from 49.10.219.164>
Feb 21 09:43:08 mgmt.team7.isucdc.com sshd[1191607]: Disconnecting authenticating user cdc 49.10.219.164 port 35554: Too many a>
Feb 21 09:43:08 mgmt.team7.isucdc.com sshd[1191616]: Invalid user alice from 49.10.219.164 port 35570
Feb 21 09:43:08 mgmt.team7.isucdc.com sshd[1191616]: pam_unix(sshd:auth): check pass; user unknown
Feb 21 09:43:08 mgmt.team7.isucdc.com sshd[1191616]: pam_unix(sshd:auth): authentication failure; logname= uid=0 euid=0 tty=ssh>
Feb 21 09:43:10 mgmt.team7.isucdc.com sshd[1191616]: Failed password for invalid user alice from 49.10.219.164 port 35570 ssh2
```

This is the ban data from fail2ban

```
[cdc@mgmt ~]$ sudo cat /root/.ssh/authorized_keys
cat: /root/.ssh/authorized_keys: No such file or directory
[cdc@mgmt ~]$ _
```

No evidence of MGMT box compromise… No evidence of access during DB_flag taken timeframe… suggesting that SQL has to be the answer

```
no crontab for root
[cdc@mgmt ~]$ sudo ls /etc/cron.d
0hourly  raid-check
[cdc@mgmt ~]$ sudo cat /etc/cron.*/* 2>/dev/null
# Run the hourly jobs
SHELL=/bin/bash
PATH=/sbin:/bin:/usr/sbin:/usr/bin
MAILTO=root
01 * * * * root run-parts /etc/cron.hourly
#!/bin/sh

/usr/sbin/logrotate /etc/logrotate.conf
EXITVALUE=$?
if [ $EXITVALUE != 0 ]; then
    /usr/bin/logger -t logrotate "ALERT exited abnormally with [$EXITVALUE]"
fi
exit $EXITVALUE
# Run system wide raid-check once a week on Sunday at 1am by default
0 1 * * Sun root /usr/sbin/raid-check

#!/bin/sh
# Check whether 0anacron was run today already
if test -r /var/spool/anacron/cron.daily; then
    day=`cat /var/spool/anacron/cron.daily`
fi
if [ `date +%Y%m%d` = "$day" ]; then
    exit 0
fi

# Do not run jobs when on battery power
online=1
for psupply in AC ADP0 ; do
    sysfile="/sys/class/power_supply/$psupply/online"

    if [ -f $sysfile ] ; then
        if [ `cat $sysfile 2>/dev/null`x = 1x ]; then
            online=1
            break
        else
            online=0
        fi
    fi
done
if [ $online = 0 ]; then
    exit 0
fi
/usr/sbin/anacron -s
[cdc@mgmt ~]$
```

proving no cron job or attacker persistence

ADMIN:

We havent seen anything abnormal on this machine to believe it has been broken into. There is nothing special on wireshark, waf, or wazuh for this machine

```
cdc@Ubuntu-External:~$ sudo nmap -sS -p- -Pn admin.t
[sudo] password for cdc:
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-2
Nmap scan report for admin.team7.isucdc.com (49.49.3
Host is up (0.0014s latency).
Not shown: 65533 filtered ports
PORT     STATE SERVICE
80/tcp   open  http
3389/tcp open  ms-wbt-server
MAC Address: BC:24:11:E1:54:D1 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 104.5
```

```
cdc@WAF:~/waf_logs$ sudo nmap -sS -Pn -p- 192.168.1.
[sudo] password for cdc:
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-2
Nmap scan report for 192.168.1.40
Host is up (0.00087s latency).
Not shown: 65533 filtered ports
PORT     STATE SERVICE
80/tcp   open  http
3389/tcp open  ms-wbt-server
MAC Address: BC:24:11:95:96:B8 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 104.3
```

The nmap ports are the same and nothing has changed…

WWW:

We havent seen anything abnormal on this machine to believe it has been broken into. There is nothing special on wireshark, waf, or wazuh for this machine

```
cdc@Ubuntu-External:~$ sudo nmap -sS -p- -Pn www.tea
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-2
Nmap scan report for www.team7.isucdc.com (49.49.33.
Host is up (0.00091s latency).
Not shown: 65533 filtered ports
PORT   STATE SERVICE
22/tcp open  ssh
80/tcp open  http
MAC Address: BC:24:11:E1:54:D1 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 104.4
```

```
cdc@WAF:~/waf_logs$ sudo nmap -sS -Pn -p- 192.168.1.
Starting Nmap 7.80 ( https://nmap.org ) at 2026-02-2
Nmap scan report for 192.168.1.50
Host is up (0.00057s latency).
Not shown: 65533 filtered ports
PORT   STATE SERVICE
22/tcp open  ssh
80/tcp open  http
MAC Address: BC:24:11:25:7D:A6 (Unknown)

Nmap done: 1 IP address (1 host up) scanned in 104.4
```

The nmap ports are the same and nothing has changed…